

Ethereum Rollup Settlement Take Rate After Dencun

Validated working paper source for the L2-to-L1 rent analysis
release candidate

autonomous-agentic-research-swarm

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1 Abstract

This paper documents the current release-candidate estimate of Ethereum rollup Settlement Take Rate (STR), the share of rollup fee revenue paid back to Ethereum L1 as settlement rent. The evidence bundle uses validated local artifacts as of 2026-04-09 and spans 14 rollups over 1,559 daily observations from 2022-01-01 through 2026-04-08 ([autonomous-agentic-research-swarm repository 2026e, 2026a](#)). On the locked regime table, mean ecosystem STR is 69.14% before Dencun and 11.68% after Dencun, with a lower 8.98% mean during post-Dencun blob-fee-floor runs ([autonomous-agentic-research-swarm repository 2026f, 2026i](#)). The manuscript is intentionally downstream of repo contracts and validation outputs: it reports the current validated surface, documents provenance, and records remaining release caveats instead of redefining metrics in prose.

2 Research Question

The repository’s locked project question is whether Ethereum L1’s settlement take rate on rollup economics is rising, stable, or decaying as L2 usage scales ([autonomous-agentic-research-swarm repository 2026e](#)). The primary metric is Settlement Take Rate (STR), defined in the protocol as

$$\text{STR}_t = (\text{sum}_i \text{RentPaid}_{\{i,t\}}) / (\text{sum}_i \text{L2Fees}_{\{i,t\}})$$

where `RentPaid` is the authoritative on-chain estimate of rollup payments to Ethereum L1 and `L2Fees` comes from the locked vendor denominator surface ([autonomous-agentic-research-swarm repository 2026f](#)). The current validated evidence supports a specific descriptive answer: ecosystem STR decays sharply after the Dencun boundary on 2024-03-13, even though rollup fee activity remains substantial ([autonomous-agentic-research-swarm repository 2026f, 2026i](#)).

3 Data And Protocol

This draft uses only durable local artifacts already produced upstream of the paper layer. The core empirical inputs are the canonical rollup panel, the daily L1 rent decomposition, the validation bundle, and the W6 release figures/table ([autonomous-agentic-research-swarm repository 2026a, 2026c, 2026b, 2026h, 2026d, 2026i](#)).

Four protocol rules matter most for interpreting the paper ([autonomous-agentic-research-swarm repository 2026f](#)):

- STR is computed daily in ETH-native units.
- A panel row exists only when both `l2_fees_eth` and `rent_paid_eth` are present.
- Dates on or after 2024-03-13 are post-Dencun.
- Blob-fee-floor runs are post-Dencun stretches of at least 7 consecutive days where blob base fee stays within 1.05 x the post-Dencun minimum.

On the validated 2026-04-09 surface, the authoritative panel contains 12,434 rollup-day rows across 14 rollups and 1,559 dates, while the decomposition covers the full 2022-01-01 through 2026-04-08 daily window ([autonomous-agentic-research-swarm repository 2026a, 2026c](#)). The manuscript therefore treats the release figures and regime table as direct downstream artifacts of the locked analysis path rather than restating results from untracked calculations.

4 Validation

All three upstream validation reports currently pass on the same 2026-04-09 as-of surface ([autonomous-agentic-research-swarm repository 2026a, 2026c, 2026b](#)). The locked checks establish three distinct claims.

First, the authoritative panel and the rent-component surface satisfy schema, primary-key, and non-null requirements, and the component identities reconcile exactly to canonical `rent_paid_eth` under both tx-family and fee-class decompositions ([autonomous-agentic-research-swarm repository 2026a](#)). Second, daily summed components match `daily_l1_rent_decomposition.l1_total_rent_eth` within the locked tolerance over all 1,559 dates ([autonomous-agentic-research-swarm repository 2026c](#)). Third, canonical and vendor panels share full matched-key coverage, show no material unexplained rollup divergences, and differ by only about 0.02% in aggregate rent over the full matched slice ([autonomous-agentic-research-swarm repository 2026b](#)).

The validation bundle still carries one documented non-gating residual worth keeping visible in release prose: in 2025-05, authoritative monthly rent is 148.42 ETH versus 166.78 ETH in the vendor benchmark, a 12.37% gap on a low-rent month ([autonomous-agentic-research-swarm repository 2026b](#), [2026g](#)). Under the locked benchmark policy, that remains diagnostic context rather than grounds to overwrite canonical rent or block the paper source.

5 Results

Figure [Figure 1](#) shows the ecosystem time series built from the validated panel surface. The series makes the main qualitative result easy to see: pre-Dencun STR is frequently elevated, while post-Dencun STR is structurally lower even when L2 fees remain meaningful ([autonomous-agentic-research-swarm repository 2026h](#)). The accompanying regime table quantifies that break. Mean ecosystem STR falls from 69.14% before Dencun to 11.68% after Dencun, while mean daily rent drops from 147.014 ETH to 19.064 ETH ([autonomous-agentic-research-swarm repository 2026i](#)).

Figure [Figure 2](#) focuses on the post-Dencun sample and shades the protocol-defined blob-fee-floor runs ([autonomous-agentic-research-swarm repository 2026f](#), [2026d](#)). The table shows that mean STR is 8.98% during blob-fee-floor periods versus 12.77% in the rest of the post-Dencun sample ([autonomous-agentic-research-swarm repository 2026i](#)). On the current validated surface, the simplest answer to the project’s primary question is therefore not “rising” or “stable” but “decaying after the Dencun regime change,” with the cheapest blob-fee conditions coinciding with the lowest observed ecosystem rent take rates.

5.1 Regime Summary Table

The locked W6 regime table is included directly from the durable table surface so the manuscript and release bundle reference the same artifact.

Daily means are ETH/day. STR columns are percentages. Blob-fee values are in gwei.

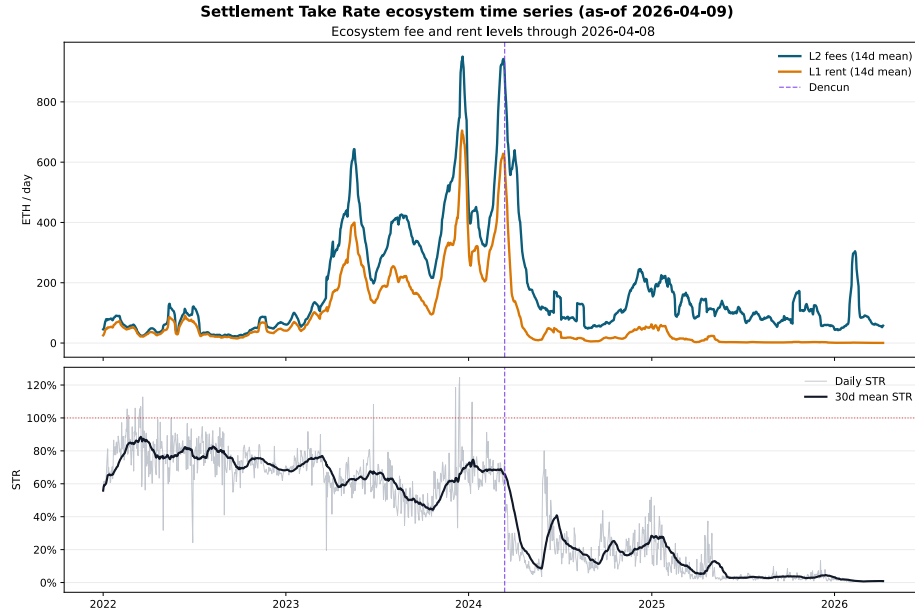


Figure 1: Settlement Take Rate ecosystem time series through 2026-04-08. The released figure overlays daily STR with smoothed L2 fee and L1 rent levels, and marks the Dencun boundary at 2024-03-13.

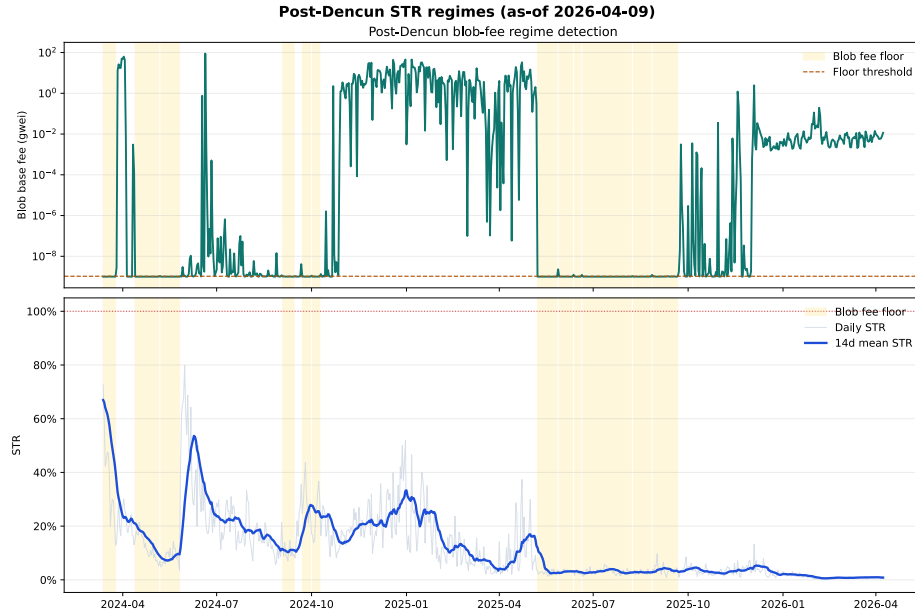


Figure 2: Post-Dencun STR regime figure with blob-fee-floor runs shaded according to the locked protocol rule.

Table 1A. Regime coverage and central tendency

Regime	Window	Days	Mean fees	Mean rent	Mean STR
Full sample	2022-01-01 to 2026-04-08	1,559	181.399	84.886	41.24%
Pre-Dencun	2022-01-01 to 2024-03-12	802	229.556	147.014	69.14%
Post-Dencun	2024-03-13 to 2026-04-08	757	130.379	19.064	11.68%
Blob fee floor	2024-03-13 to 2025-09-21	217	128.737	18.740	8.98%
Post-Dencun ex floor	2024-03-26 to 2026-04-08	540	131.038	19.194	12.77%

Table 1B. STR distribution and blob-fee diagnostics

Regime	Median STR	P90 STR	Mean blob fee
Full sample	44.59%	79.85%	2.250096
Pre-Dencun	69.36%	84.97%	n/a
Post-Dencun	6.55%	27.64%	2.250096
Blob fee floor	3.93%	24.06%	0.000000
Post-Dencun ex floor	9.09%	28.61%	3.154302

6 Provenance And Limitations

This manuscript is downstream of the locked repository control plane, not a parallel analysis channel. Protocol definitions come from `docs/protocol.md`, project-level release requirements come from `contracts/project.yaml`, and the empirical claims quoted here are limited to validated report artifacts and their provenance manifests ([autonomous-agentic-research-swarm repository 2026f](#), [2026e](#), [2026a](#), [2026c](#)).

Two limitations matter for interpretation. First, the authoritative panel is intentionally denominator-keyed: rollup-days missing vendor fee denominators

are omitted from STR aggregation even when rent-component data exists. The current validated surface therefore has 12,434 panel rows but 12,563 rent-component rows, and the component-to-decomposition identity is the correct full-rent coherence check ([autonomous-agentic-research-swarm repository 2026a, 2026g](#)). Second, this task owns manuscript source only. Final rendered artifacts, release manifests, and catalog updates remain Operator-owned T080 surfaces.

Within those limits, the paper source is release-aligned: it uses the locked filenames, cites only durable local artifacts, and can be rendered by Quarto without inventing new evidence or changing scientific definitions.

autonomous-agentic-research-swarm repository. 2026a. *Canonical Rollup Panel Validation*.

autonomous-agentic-research-swarm repository. 2026b. *Cross-Source Reconciliation*.

autonomous-agentic-research-swarm repository. 2026c. *L1 Rent Decomposition Validation*.

autonomous-agentic-research-swarm repository. 2026d. *Post-Dencun STR Regimes*.

autonomous-agentic-research-swarm repository. 2026e. *Project Contract: L2-to-L1 Rent Analysis*.

autonomous-agentic-research-swarm repository. 2026f. *Protocol Lock (Phase 0)*.

autonomous-agentic-research-swarm repository. 2026g. *Release Output Caveats for T070*.

autonomous-agentic-research-swarm repository. 2026h. *Settlement Take Rate Ecosystem Time Series*.

autonomous-agentic-research-swarm repository. 2026i. *STR Regime Summary Table*.